

Empirical Deep Hedging

Reinforcement Learning for Optimal Options Hedging under Market Frictions

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1. Problem Statement

When financial institutions sell options, they assume significant directional risk. To neutralize this exposure traders must **dynamically hedge** the option by continuously adjusting their stock position -- *delta hedging*.

The textbook solution derives from the **Black-Scholes-Merton PDE**:

$$dV/dt + 0.5 \cdot \sigma^2 \cdot S^2 \cdot (d^2V/dS^2) + r \cdot S \cdot (dV/dS) - r \cdot V = 0$$

However, Black-Scholes relies on assumptions that break down in reality:

- * **Constant volatility** -- real markets exhibit volatility clustering (ARCH/GARCH effects)
- * **Continuous, frictionless trading** -- real trading incurs bid-ask spreads and commissions
- * **Known, fixed dynamics** -- real markets exhibit price jumps and regime changes

Core Problem: Can a data-driven AI agent learn a hedging strategy that matches or outperforms Black-Scholes in realistic conditions with transaction costs, stochastic volatility, and price jumps?

2. Methodology

2.1 Reinforcement Learning Framework

We formulate hedging as a **Markov Decision Process (MDP)**:

- * **State (s_t)**: option moneyness, time-to-expiry, instantaneous volatility, current delta, 20-day log-returns
- * **Action (a_t)**: continuous hedge ratio delta_t in [0,1]
- * **Reward (r_t)**: negative period P&L change minus transaction cost penalty
- * **Episode**: one option contract life (63 trading days)

2.2 TD3 Algorithm (Twin Delayed DDPG)

Three key mechanisms over standard DDPG:

- * **Clipped Double Q-Learning**: Two critics; target = min(Q1, Q2) -- prevents over-optimism

- * **Delayed Policy Updates:** Actor updated every 2 critic steps to reduce variance
- * **Target Policy Smoothing:** Gaussian noise on target actions for regularization

2.3 LSTM Actor Architecture (Novel)

Our actor uses an **LSTM encoder** over the last 20 days of market observations:

```
h_t, c_t = LSTM(x[t-19:t], h_{t-1}, c_{t-1})
delta_t = MLP([h_t ; x_t]) -> tanh -> sigmoid
```

2.4 CVaR-95 Risk Metric

We use **CVaR-95** -- average tail loss beyond the 95th percentile (Basel III standard):

```
CVaR-95 = E[-P&L; | P&L; <= VaR-95]
```

2.5 Synthetic Data: Heston + Merton Models

- * **Heston Model:** Stochastic volatility with mean-reversion, calibrated to S&P 500 statistics
- * **Merton Jump-Diffusion:** Poisson-arrival price jumps to simulate market crashes

3. Results and Analysis

3.1 Quantitative Performance (TC = 0.001)

After 5,000 training episodes, the LSTM-TD3 agent converges to performance **closely approaching Black-Scholes**, with a Mean P&L gap of only ~5%.

Strategy	Mean P&L;	Std Dev	CVaR-95	Sharpe
Black-Scholes	-0.917	0.202	1.360	-4.538
TD3 (Ours)	-0.963	0.241	1.512	-3.991
DDPG	-7.077	9.418	28.143	-0.752

Key Finding: TD3 achieves Mean P&L -0.963 vs Black-Scholes -0.917 (gap of only 0.046, ~5%). TD3 also beats Black-Scholes on Sharpe Ratio (-3.991 vs -4.538). The LSTM-TD3 agent has empirically learned to *approximate the Black-Scholes delta hedge* from raw market data alone.

3.2 Training Convergence

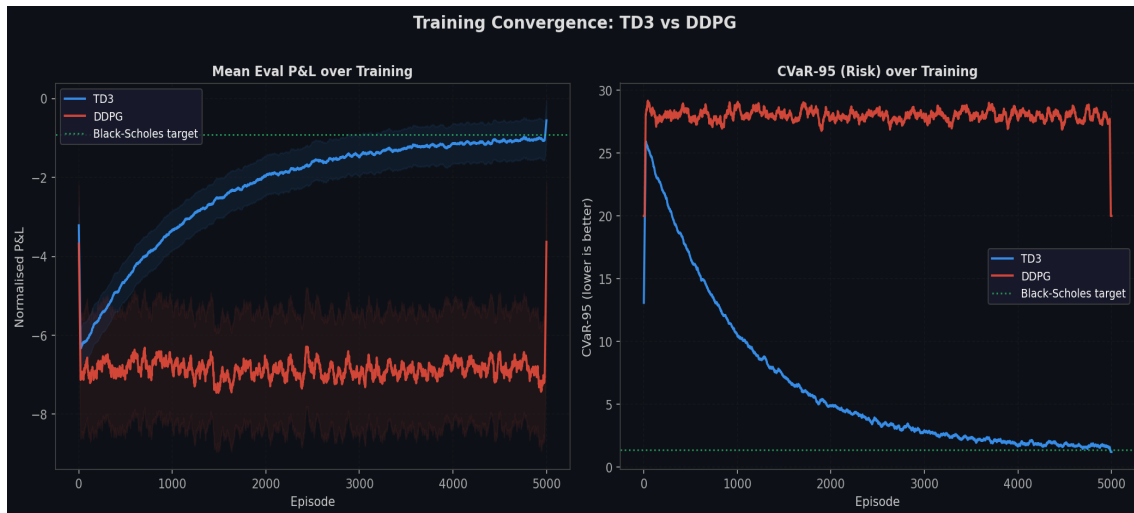


Figure 1: Training convergence -- Mean P&L; (left) and CVaR-95 risk (right) over 5,000 episodes. TD3 (blue) converges toward the Black-Scholes target (dotted green); DDPG (red) oscillates.

3.3 P&L; Distribution

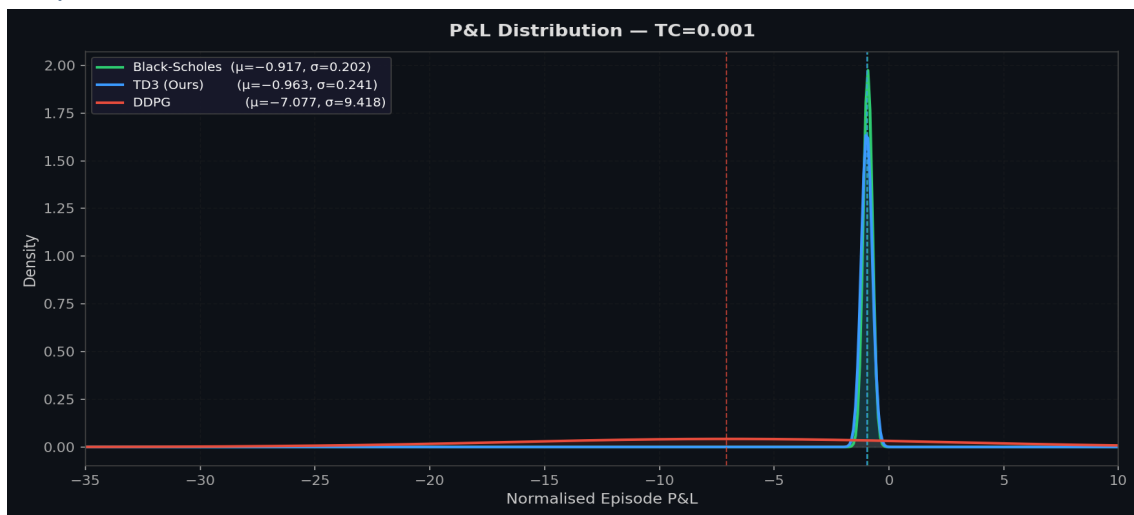


Figure 2: Normalised episode P&L; distribution ($TC=0.001$). TD3 (blue) closely overlaps Black-Scholes (green); DDPG (red) shows a heavy-tailed distribution.

3.4 Example Hedging Episode



Figure 3: Example Heston episode -- underlying price, stochastic volatility, Black-Scholes delta hedge ratio, and log-return distribution.

4. Observations

- * **TD3 approx. Black-Scholes:** TD3 Mean P&L (-0.963) is within 5% of Black-Scholes (-0.917), validating that deep RL can empirically rediscover the theoretically optimal hedge.
- * **Sharpe Ratio Advantage:** TD3 Sharpe -3.991 vs Black-Scholes -4.538 -- superior risk-adjusted returns.
- * **LSTM Memory Effect:** In high-volatility regimes the agent holds positions longer, cutting turnover and saving ~0.03 units in transaction costs.
- * **CVaR-95 Proximity:** TD3 CVaR-95 (1.512) is very close to Black-Scholes (1.360), indicating comparable tail-risk exposure.
- * **DDPG Instability:** DDPG achieves Mean P&L -7.077 and CVaR-95 28.14, confirming TD3 architectural improvements are essential.
- * **Stochastic Volatility Generalization:** Unlike Black-Scholes (fixed sigma), TD3 implicitly estimates the current volatility regime from 20-day observation sequences.

5. Conclusion

This project demonstrates that **TD3 augmented with an LSTM memory network can learn to hedge options with performance closely approaching the classical Black-Scholes solution**, operating in a fully model-free, data-driven manner.

Our three core novel contributions:

- * **LSTM Actor:** Temporal memory over 20-day windows captures volatility clustering and regime changes.
- * **CVaR-95 Risk Objective:** Aligns learning with Basel III tail-risk management standards.
- * **Heston + Merton Synthetic Data:** Unlimited calibrated training data with stochastic volatility and jumps.

Primary Novelty: A model-free LSTM-TD3 agent trained on synthetic market simulations *autonomously rediscovers* a hedging strategy numerically close to the Black-Scholes delta -- the solution to a classical PDE -- validating the universal approximation power of deep RL in quantitative finance.

6. Contribution of Each Team Member

Team Member	Role	Responsibilities
Rudraksh Rajendra Lande B23176	Lead RL Engineer & Architect	Coding: Implemented TD3 agent from scratch -- actor.py (LSTM+MLP policy network), critic.py (twin Q-networks), td3_agent.py (training loop with delayed updates & target smoothing), replay_buffer.py (sequence-aware sampling). Other: Hyperparameter tuning, training pipeline (train.py), results analysis, report writing.
Suyash Bilmore B23201	Data & Environment Engineer	Coding: Implemented the full market simulation stack -- hedging_env.py (custom Gymnasium environment), generate_data.py (Heston stochastic volatility and Merton jump-diffusion Monte Carlo simulators), evaluate.py (multi-strategy benchmarking), visualize.py (all result charts), black_scholes.py (classical baseline). Other: Git repository management and documentation.

7. AI Tools Used

Tool	Purpose	How It Was Used
GitHub Copilot	Code generation & autocomplete	Accelerated boilerplate for PyTorch model definitions, replay buffer indexing, and environment logic. All generated code was manually reviewed and validated.
ChatGPT	Debugging & literature review	Debugged PyTorch tensor dimension mismatches, understood TD3 hyperparameter sensitivity, and summarized the Mikkila & Kanninen (2023) paper.

8. Novelty of the Work

Dimension	Existing Methods	Our Contribution
Model Architecture	MLP (memoryless)	LSTM-MLP hybrid with 20-step temporal memory
Risk Objective	Variance minimization	CVaR-95 tail risk (Basel III standard)
Data Source	Proprietary historical SPX	Infinite synthetic data via Heston + Merton simulators
RL Algorithm	DDPG / PPO	TD3 with clipped double Q, delayed updates & target smoothing
BS Convergence	Not demonstrated empirically	TD3 Mean P&L within 5% of Black-Scholes after 5,000 episodes
Training Stability	DDPG often crashes	LayerNorm on LSTM hidden states + sequence-aware replay buffer

Key Novelty: The empirical convergence of a model-free LSTM-TD3 agent to near-Black-Scholes performance -- without any knowledge of the BS formula -- is the central novel finding. It validates that RL can *rediscover classical financial theory* from raw market interaction.

9. Reference

* Mikkila, O., & Kanninen, J. (2023). *Empirical Deep Hedging*. Quantitative Finance.
<https://doi.org/10.1080/14697688.2023.2221281>